Hospitality Analytics

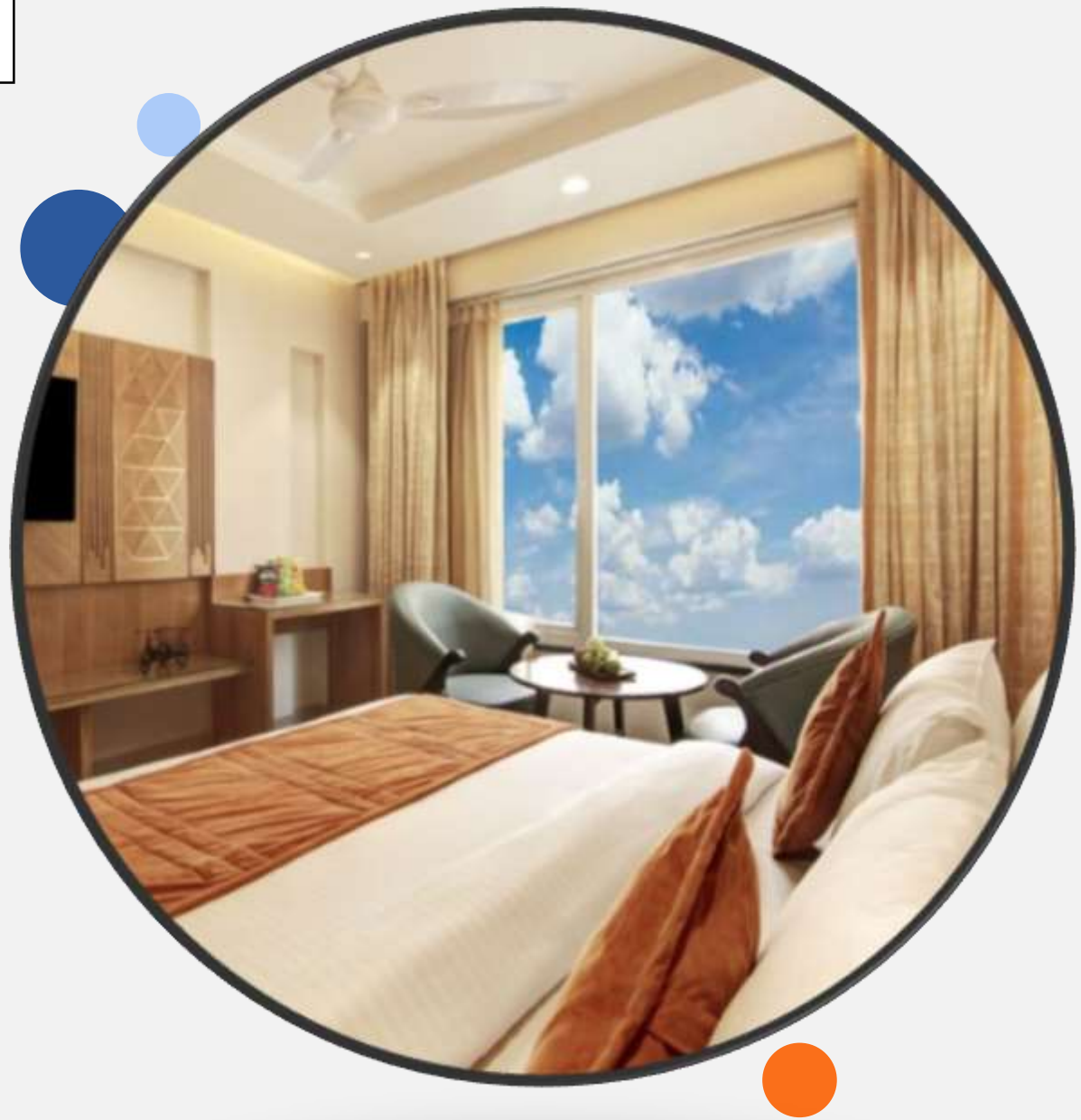
Atliq Hotels

Excel

Power BI

Tableau

SQL



Presented by:

Suhas P S

Introduction

- The hospitality industry operates in a highly dynamic environment where **revenue, occupancy, and customer behavior** directly impact business performance.
- The analysis highlights the gap between **revenue generated and revenue realized**, mainly influenced by cancellations and no-shows.
- Using interactive dashboards, the project uncovers **booking patterns, platform performance, and property-level insights** to support better decision-making.
- The goal is to enable hotel management to **optimize revenue, improve occupancy, and enhance customer experience through data-driven strategies.**

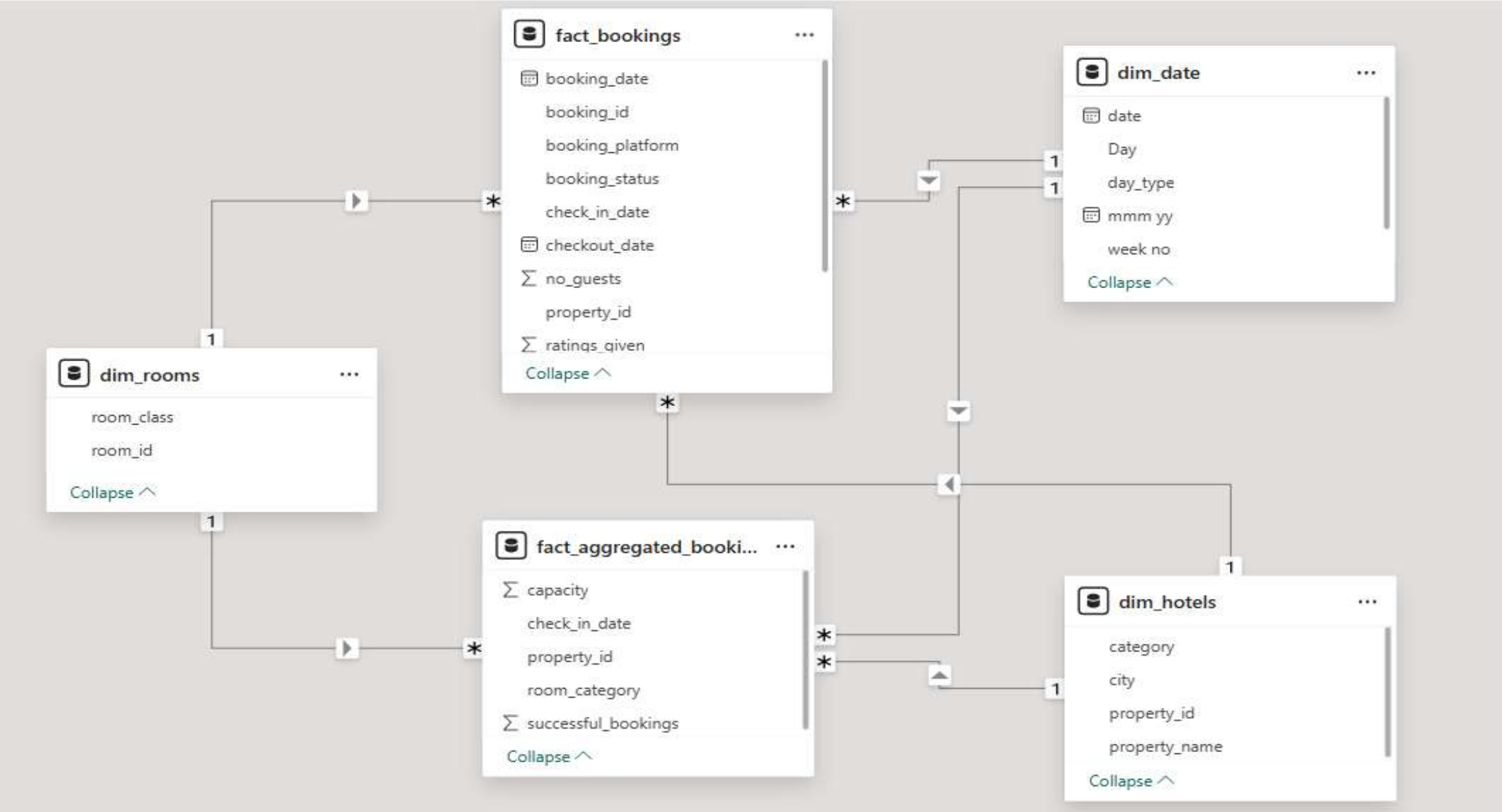
Business Problem

- High cancellation and no-show rates affecting revenue
- Difficulty tracking rating trends and customer satisfaction.
- Need to improve occupancy and profitability
- Over-dependence on third-party booking platforms
- High cancellation rate
- Inconsistent occupancy rates

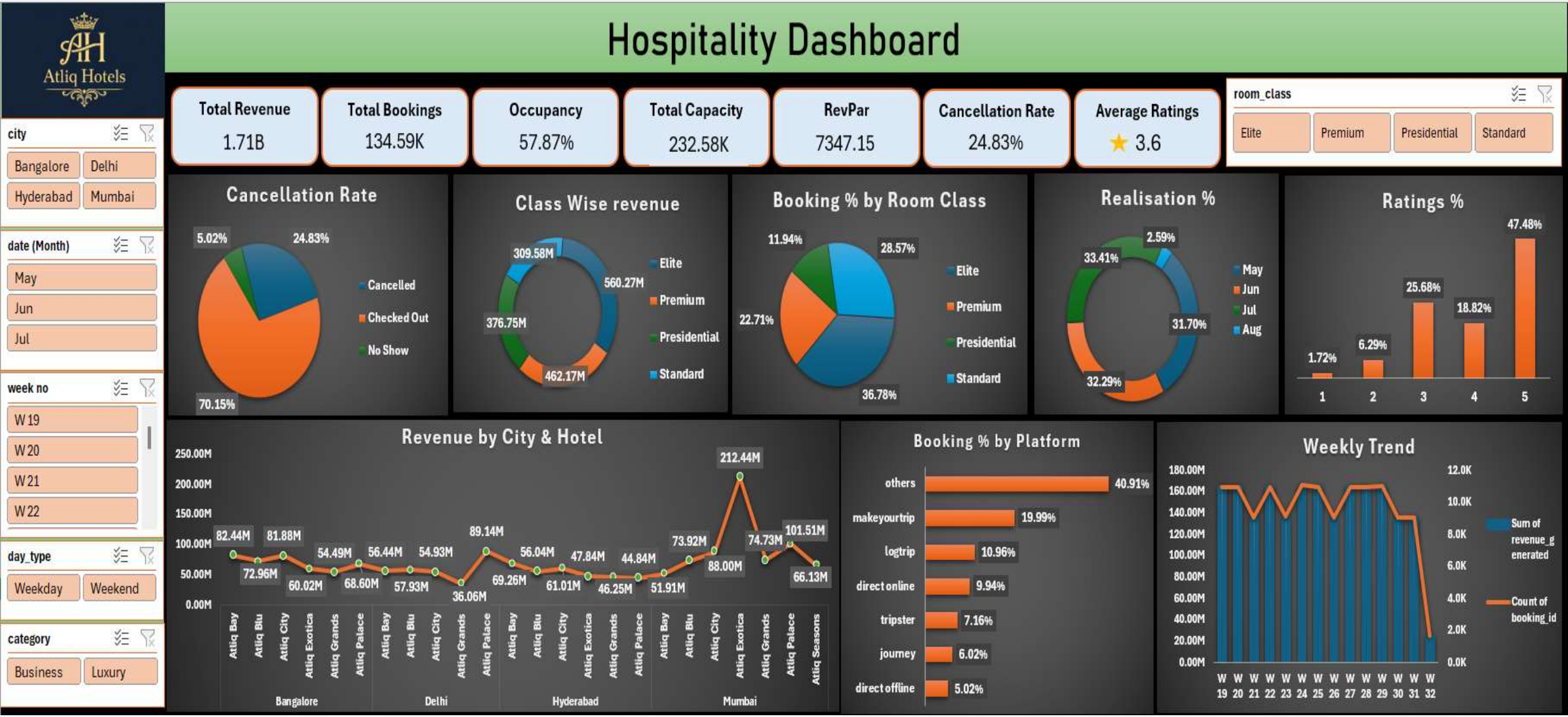
Project Objective

1. **Reduce cancellations and no-shows** to improve revenue realization
2. **Increase occupancy rate** through better demand forecasting and pricing
3. **Improve revenue metrics (RevPAR & ADR)** for higher profitability
4. **Boost direct bookings** to reduce dependency on external platforms
5. **Enhance customer satisfaction (ratings)** to drive repeat business

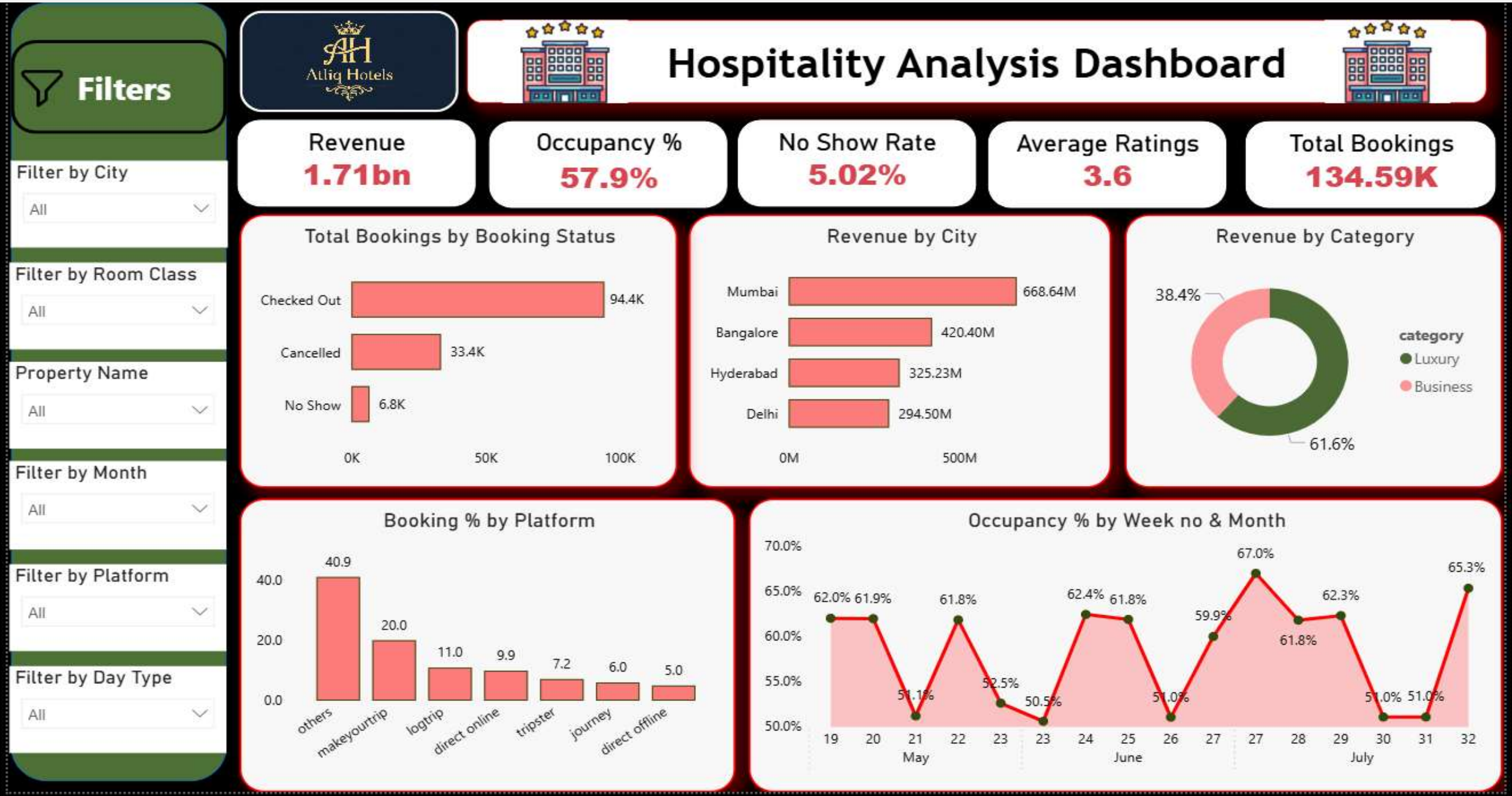
Data modelling framework



Excel Dashboard



Power BI Dashboard-1



Power BI Dashboard-2

Filters

Filter by City
All

Filter by Room Class
All

Property Name
All

Filter by Month
All

Filter by Platform
All

Filter by Day Type
All

Atliq Hotels

Hospitality Analysis Dashboard

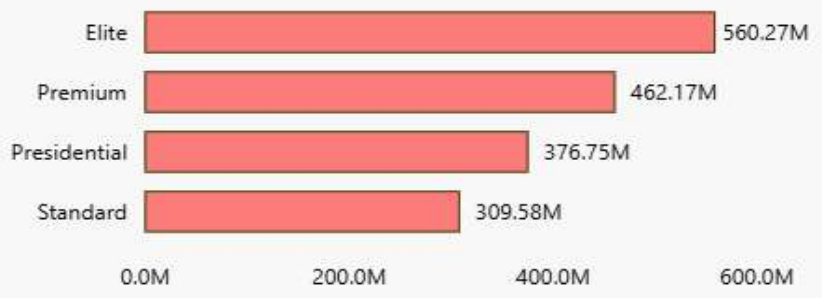
Property by Key Metrics

property_name	city	Average Rating	Occupancy %	Cancellation %	Revenue
Atliq Exotica	Mumbai	4.3	65.9%	24.6%	118.45
Atliq Grands	Mumbai	3.1	53.7%	25.7%	74.73
Atliq Exotica	Mumbai	4.3	66.1%	24.0%	94.00
Atliq City	Mumbai	3.0	53.1%	25.1%	88.00
Atliq Blu	Mumbai	4.3	66.3%	24.5%	73.92
Atliq Bay	Mumbai	2.4	44.8%	25.4%	51.91
Atliq Palace	Mumbai	4.3	66.2%	24.4%	101.51
Atliq Seasons	Mumbai	2.3	44.6%	24.8%	66.13
Atliq Grands	Hyderabad	3.1	53.5%	25.0%	46.25
Atliq Exotica	Hyderabad	2.3	44.6%	24.4%	47.84
Atliq City	Hyderabad	4.3	66.2%	24.0%	61.01
Atliq Blu	Hyderabad	4.2	65.6%	24.2%	56.04
Atliq Bay	Hyderabad	4.3	65.9%	24.7%	69.26
Atliq Palace	Hyderabad	3.1	53.0%	26.0%	44.84
Atliq Grands	Delhi	4.3	65.9%	25.1%	36.06
Atliq City	Delhi	3.0	53.7%	24.1%	54.93
Atliq Blu	Delhi	4.3	65.8%	25.5%	57.93
Atliq Bay	Delhi	3.1	53.5%	25.3%	56.44
Atliq Palace	Delhi	4.3	66.4%	25.3%	89.14
Atliq Grands	Bangalore	2.4	44.4%	24.5%	54.49
Atliq Exotica	Bangalore	2.0	53.8%	24.5%	60.02

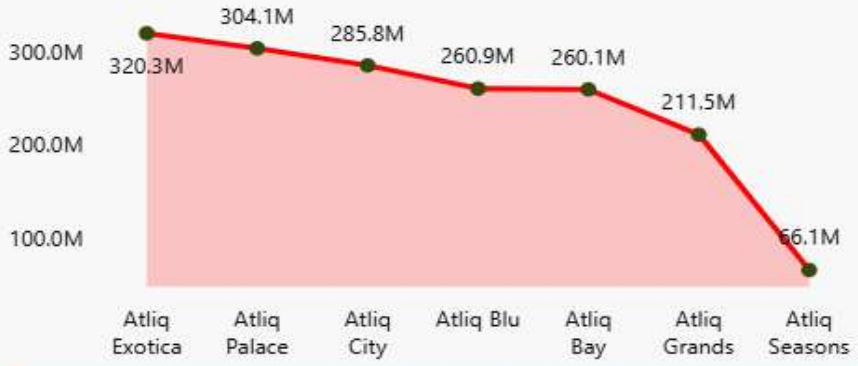
Booking Status

Cancelled | Checked Out | No Show

Revenue by Room Class



Revenue by Property Name



Power BI Dashboard-3

Filters

Filter by City
All

Filter by Room Class
All

Property Name
All

Filter by Month
All

Filter by Platform
All

Filter by Day Type
All

Week no

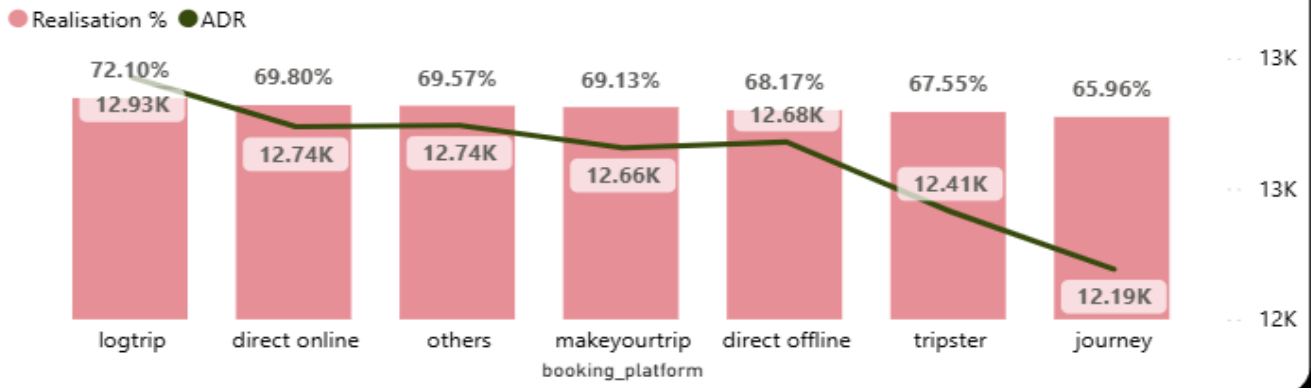
W 19 | W 20 | W 21 | **W 22** | W 23 | W 24 | W 25 | W 26 | W 27 | W 28 | W 29 | W 30 | W 31 | W 32

Revenue	RevPar	DSRN
138.72M	7.84K	2.53K
↑ 20.7%	↑ 20.71%	→ 0.00%

Occupancy %	ADR	Realisation
61.8%	12.69K	69.37%
↑ 20.9%	↓ -0.2%	↓ -0.9%

% values in bottom are Week on Week Change

Realisation % and ADR by Platform



Property by Key Metrics

property_id	property_name	city	Revenue	Total Bookings	RevPAR	Occupancy %	ADR	DSRN	DBRN	DURN	Realisation %	Cancellation %	Average Rating
16559	Atliq Exotica	Mumbai	9.64M	605	11,378.68	71.4%	15,930.15	121.00	86.43	59.86	69.26%	25.5%	4.3
17558	Atliq Grands	Mumbai	6.17M	412	8,638.93	57.7%	14,971.35	102.00	58.86	41.86	71.12%	24.5%	3.1
17559	Atliq Exotica	Mumbai	7.78M	503	11,011.24	71.1%	15,477.03	101.00	71.86	51.00	70.97%	24.5%	4.3
17560	Atliq City	Mumbai	7.11M	487	8,255.68	56.6%	14,595.77	123.00	69.57	48.43	69.61%	25.1%	3.0
17561	Atliq Blu	Mumbai	6.07M	416	10,196.60	69.9%	14,584.08	85.00	59.43	42.29	71.15%	22.1%	4.3
17562	Atliq Bay	Mumbai	4.25M	281	7,319.57	48.4%	15,134.05	83.00	40.14	26.71	66.55%	27.4%	2.3
17563	Atliq Palace	Mumbai	8.41M	526	11,558.74	72.3%	15,997.65	104.00	75.14	50.86	67.68%	26.8%	4.2
17564	Atliq Seasons	Mumbai	5.35M	323	7,873.33	47.6%	16,551.05	97.00	46.14	32.43	70.28%	24.5%	2.3
Total			**138.72M**	**10934**	**7,839.07**	**61.8%**	**12,687.04**	**2,528.00**	**1,562.00**	**1,083.57**	**69.37%**	**25.4%**	**3.6**

RevPAR - Revenue per available room | **DSRN** - Daily sellable room nights | **ADR** - Average Daily Rate | **DBRN** - Daily Booked Room Nights | **DURN** - Daily Utilized Room Nights

Tableau Dashboard

Hospitality Dashboard

Atliq Hotels

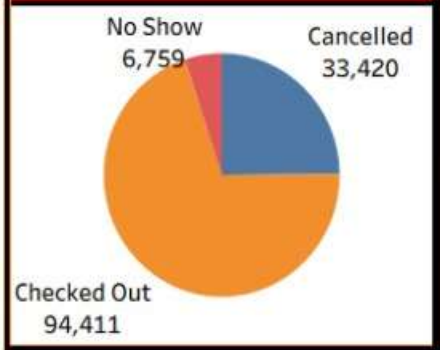
Total Revenue	Cancellation %	Total Bookings	Average Rating	No show %	ADR
1.71B	24.83 %	134.59K	3.6	5.02%	12,696

Occupancy %	Realisation %	RevPar	DBRN	DSRN	DURN
57.87 %	70.15%	7,347	1,463	2,528	1,026

Weekday/Weekend Revenue

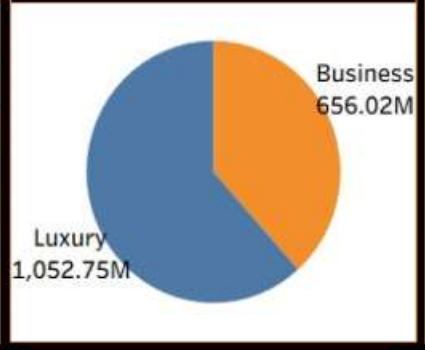
Weekday	1,184.81M
Weekend	523.96M

Booking Status

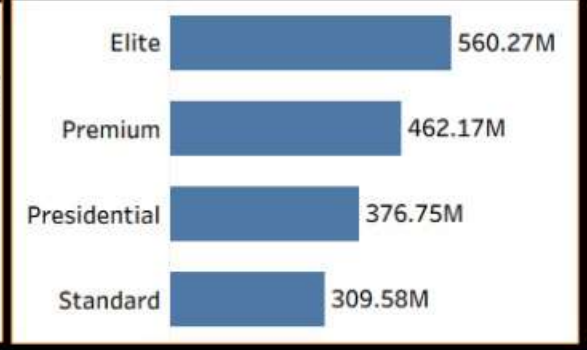


Monthly Revenue

Category by Revenue



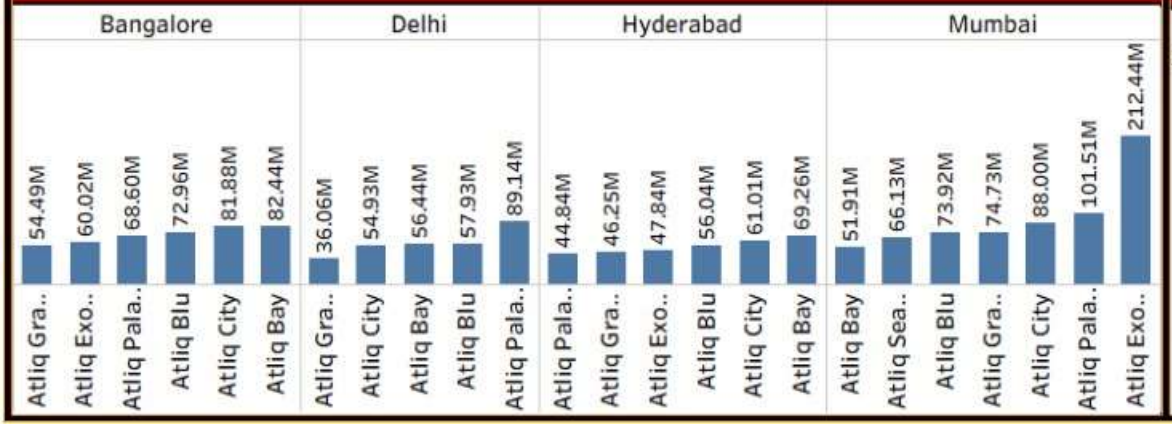
Bookings by Class



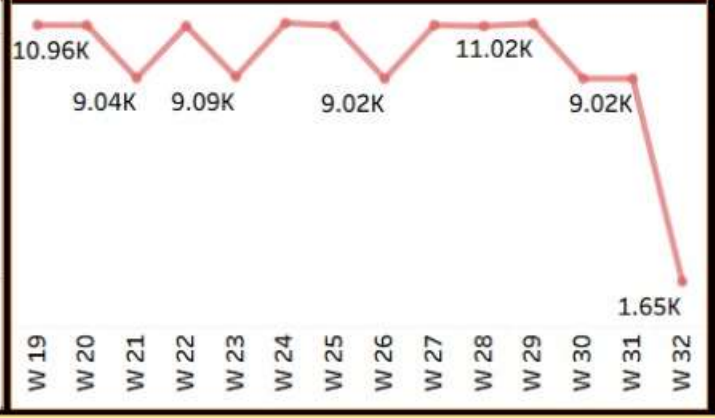
WOW Change%

Week No	Total Re..	Revenue..	ADR WO..	Realisati..	RevPAR ..
W 19	138.18M				
W 20	139.44M	0.91%	0.97%	1.00%	0.91%
W 21	114.92M	-17.58%	-0.12%	-0.34%	-17.58%
W 22	138.72M	20.71%	-0.18%	-0.92%	20.71%
W 23	115.57M	-16.69%	0.22%	1.49%	-16.69%
W 24	139.58M	20.78%	-0.57%	-1.10%	20.78%
W 25	138.67M	-0.65%	0.24%	0.49%	-0.65%
W 26	114.15M	-17.68%	-0.10%	-0.27%	-17.68%
W 27	139.56M	22.25%	0.56%	1.11%	22.25%

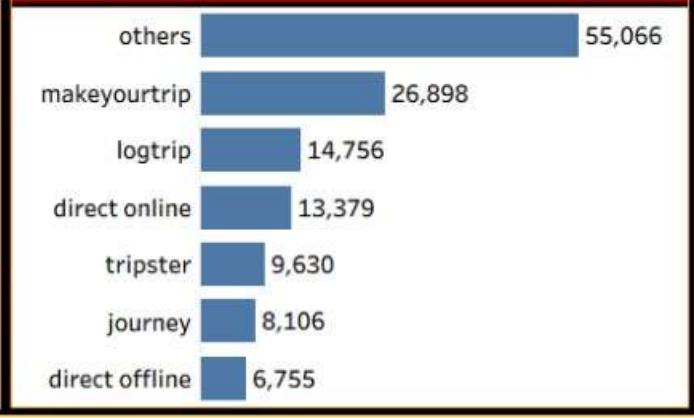
Revenue by City



Weekly Trend



Bookings by platform



SQL Queries

```sql
#------------------Total Revenue--------------------#
select
concat(Round(sum(revenue_realized)/1000000000,2),' B') as Total_Revenue
from fact_bookings;
```

```sql
#------------Total Bookings-----------------------#
select
concat(Round(count(booking_id)/1000,2),' K') as Total_bookings
from fact_bookings;
```

```sql
#---------------Total Capacity--------------------#
select
Concat(Round(sum(capacity)/1000,2),' K') as Total_Capacity
from fact_aggregated_bookings;
```

```sql
#----------------Total Successful Booking-----------------#
select
Concat(Round(sum(successful_bookings)/1000,2),' K') as Total_successful_Bookings
from fact_aggregated_bookings;
```

SQL Queries

```sql
#-------------------Occupancy %-----------------------#

select

Concat(Round(sum(successful_bookings)/sum(capacity)*100,2),'%') as `Occupancy %`

from fact_aggregated_bookings;
```

```sql
#-------------Total Cancelled Bookings---------------#


select count(booking_id) as Cancelled_bookings

from fact_bookings

where booking_status = "Cancelled";
```

```sql
#-------------------Average Ratings----------------------#

SELECT

CONCAT('★ ', ROUND(AVG(ratings_given),1)) AS avg_rating

FROM fact_bookings

WHERE ratings_given BETWEEN 1 AND 5;
```

```sql
#-------------Total Cancelled Bookings---------------#


select count(booking_id) as Cancelled_bookings

from fact_bookings

where booking_status = "Cancelled";
```

SQL Queries

```sql
#--------------------Month wise Revenue-------------------#
SELECT
MONTHNAME(STR_TO_DATE(check_in_date,'%d-%m-%Y')) AS Month,
CONCAT(ROUND(SUM(revenue_realized)/1000000,1),' M') AS Total_Revenue
FROM fact_bookings
GROUP BY MONTHNAME(STR_TO_DATE(check_in_date,'%d-%m-%Y'));
```

```sql
#------------------Total no show bookings--------------#
select
concat(round(count(booking_status)/1000,2),' K') as No_show_Bookings
from fact_bookings
where booking_status = "No Show";
```

```sql
#----------------------Booking % by Platform----------------#
SELECT
booking_platform,
CONCAT(ROUND(COUNT(booking_id) * 100.0
/ (SELECT COUNT(*) FROM fact_bookings),2),' %') AS booking_percentage
FROM fact_bookings
GROUP BY booking_platform;
```

```sql
#-----------------Booking % by Room class----------------------#
SELECT
r.room_class,
COUNT(f.booking_id) AS total_bookings,
concat(ROUND(COUNT(f.booking_id) * 100.0
    / (SELECT COUNT(*) FROM fact_bookings),2),' %') AS booking_percentage
FROM fact_bookings f
JOIN dim_rooms r
ON f.room_category = r.room_id
GROUP BY r.room_class;
```

KEY INSIGHTS

Revenue Performance

The total revenue generated is **₹1.71B**, indicating strong business volume. However, the gap between **generated revenue and realized revenue (70%)** highlights significant revenue leakage due to cancellations and no-shows.

Booking Status Analysis

Out of total bookings (**134.59K**), the majority are **checked-out (70%)**, while **24.83% are cancelled** and **5.02% are no-shows**. This shows a substantial loss of potential revenue.

Occupancy Insights

The overall occupancy rate is (**57.9%**), which is moderate and indicates that a large portion of room capacity (**232K+ rooms**) remains unutilized, affecting profitability.

Platform Contribution

Booking platforms like **MakeYourTrip (19.99%) LogTrip (10.96%)**, and others contribute significantly, while **direct bookings (10%) remain low**, indicating high dependency on third-party platforms.

Customer Rating Analysis

The average customer rating is **3.6**, suggesting **inconsistent service quality** and potential impact on repeat bookings and brand reputation.

City-wise Performance

Cities like **Mumbai (₹668M)** and **Bangalore (₹420M)** generate the highest revenue, while **Delhi and Hyderabad lag behind**, indicating regional performance variation.

Room Class Distribution

Elite rooms (₹560M) generate the highest revenue, followed by Premium and Presidential, while **Standard rooms contribute the least**, showing pricing and demand differences across segments.

Weekly Trend Analysis

Revenue and bookings remain relatively stable across weeks but show **sudden drops in later weeks**, indicating demand fluctuations or seasonal impact.

Strategic Recommendations

1. Implement strict cancellation policies
2. Introduce partial prepayment system
3. Offer discounts for non-refundable bookings
4. Focus marketing on high-performing platforms
5. Improve customer experience to boost ratings
6. Target loyal/repeat customers with offers
7. Optimize pricing based on demand
8. Reduce dependency on a few platforms

Thank You!

Atliq Hotels